

- 7 Algae are aquatic photosynthetic prototists. The alga *Alexandrium* is red in colour. *Alexandrium* causes harmful algal blooms (HABs) that turn seawater red when *Alexandrium* numbers rise to high levels. The abundance of *Alexandrium* depends mainly on abiotic factors.

Alexandrium releases a toxin that accumulates in shellfish such as oysters and mussels. Humans who eat shellfish contaminated with *Alexandrium* toxin show signs of poisoning, which in some cases may cause death.

Figure 7.1 shows the global number of cases of poisoning linked to *Alexandrium* HABs between 1986 and 2018. The line shows the long-term trend in cases of poisoning due to *Alexandrium* HABs.

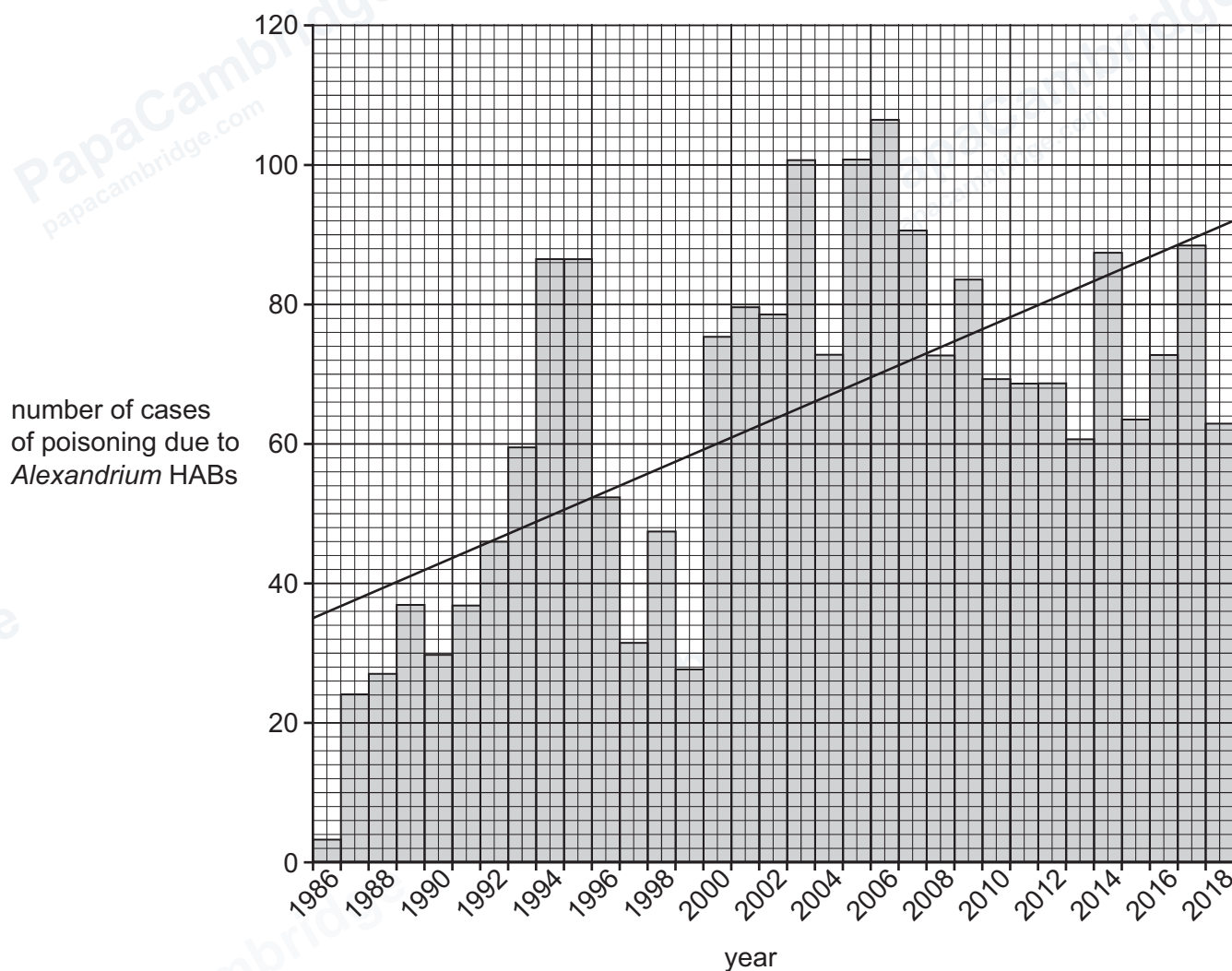
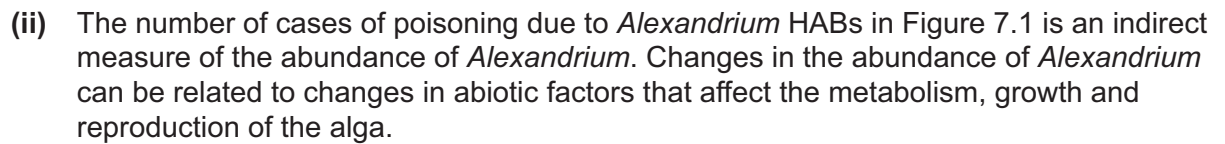


Figure 7.1

- (a) (i) Use the long-term trend line shown in Figure 7.1 to calculate the percentage change in cases of poisoning due to *Alexandrium* HABs between the start of 1986 and the end of 2018.

Show your working **and** give your answer to **three** significant figures.

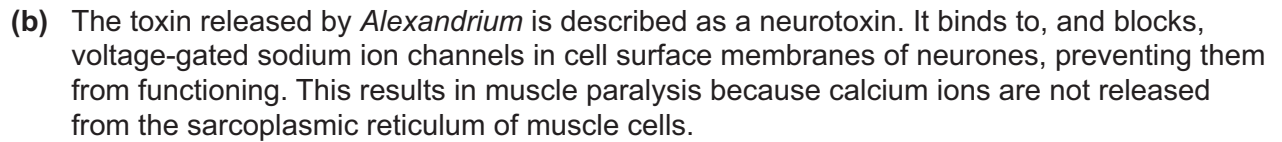
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